

Membership versus variety, and the partial-solution strata: what changes if the PDE is added to the system before decomposition

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A companion note to *A New Method of Solving PDEs*. The core algorithm *reduces* the PDE P modulo the ansatz A and projects the remainder onto the constants, returning the locus $\mathcal{V}$ of constants for which the ansatz family solves the PDE. A natural-looking alternative is to adjoin P to the ansatz and run a differential decomposition (Rosenfeld–Gröbner or, better, differential Thomas) on the *combined* system $A \cup \{P\}$. This note records why that answers a *different* question, names the discrepancy — the *partial-solution strata* — and exhibits it on a two-line example.

1 Two different loci

Let $A(c) = \{a_1, \dots, a_k\}$ be the ansatz, a differentially triangular set whose coefficients involve constant parameters $c \in \mathbb{C}^n$, and let P be the PDE. The core method computes the **membership locus**

$$\mathcal{V} = \{c \in \mathbb{C}^n : P \in [A(c)]\}, \quad (1)$$

a *universal* condition: *every* function in the ansatz family at c solves the PDE. Operationally, Ritt’s full reduction of P modulo A yields a remainder $r = \sum_{\alpha} m_{\alpha} p_{\alpha}(c)$ with the m_{α} distinct power products in the non-constant indeterminates; r vanishes *identically* iff every coefficient $p_{\alpha}(c)$ vanishes, so $\mathcal{V} = V(p_1, \dots, p_N)$.

Decomposing the combined system computes instead the **joint solution variety**

$$\text{Sol}(A) \cap \text{Sol}(P), \quad (2)$$

an *existential* condition: there *exists* a function satisfying both A and P . Projecting (2) onto the constants gives

$$\pi_c(\text{Sol}(A) \cap \text{Sol}(P)) = \mathcal{V} \cup \{\text{partial-solution strata}\} \supseteq \mathcal{V}. \quad (3)$$

Definition 1 (partial-solution stratum) *A partial-solution stratum is a set of constant values c at which a proper, nontrivial sub-family of the ansatz solutions satisfies P , while the family as a whole does not. Equivalently: $c \notin \mathcal{V}$, yet $\text{Sol}(A(c)) \cap \text{Sol}(P)$ contains more than the zero solution.*

A prerequisite is immediate from Definition 1: the ansatz solution space at c must be at least two-dimensional. A one-dimensional family either solves P in its entirety or only through its zero member; there is no proper nontrivial sub-family to peel off.

2 A two-dimensional ansatz where the strata appear

Example 1 Take a two-dimensional ansatz and a first-order PDE,

$$A(c) : \quad \Psi'' - c\Psi = 0, \quad P : \quad \Psi' - \Psi = 0. \quad (4)$$

For $c = \omega^2$ the ansatz solution space is two-dimensional, $\Psi = K_1 e^{\omega x} + K_2 e^{-\omega x}$.

Membership (core method): reduce P modulo A . The leader of P is Ψ' , which is ranked below the leader Ψ'' of A , so P is already fully reduced with respect to A ; the remainder is P itself,

$$r = \Psi' - \Psi = (+1) \cdot \Psi' + (-1) \cdot \Psi.$$

The projection coefficients are the constants $+1$ and -1 : never zero, with no c -dependence. Hence

$$\mathcal{V} = V(1, -1) = \emptyset.$$

This is correct: no value of c makes the entire two-parameter family $\{K_1 e^{\omega x} + K_2 e^{-\omega x}\}$ obey the first-order condition $\Psi' = \Psi$.

Combined decomposition: adjoin P , rank the constant lowest. The decomposition selects the lower-leader equation $\Psi' - \Psi$ into the triangular set first, then reduces $\Psi'' - c\Psi$ against it. Using $\Psi' = \Psi \Rightarrow \Psi'' = \Psi$,

$$\Psi'' - c\Psi \longrightarrow \Psi - c\Psi = (1 - c)\Psi = 0.$$

This equation has leader Ψ and initial $(1 - c)$, so the algorithm splits on the vanishing of $(1 - c)$:

cell	reduced system	solutions
$c \neq 1$	$(1 - c)\Psi = 0, \quad \Psi' = \Psi$	$\Psi \equiv 0$ (trivial only)
$c = 1$	$0 = 0, \quad \Psi' = \Psi$	$\Psi = K e^x$ (one-parameter)

The cell $c = 1$ is the partial-solution stratum. At $c = 1$ the ansatz has the two modes e^x and e^{-x} ; the PDE $\Psi' = \Psi$ retains the e^x mode and discards e^{-x} . Exactly half the family solves the PDE — “some, but not all” — so $c = 1 \notin \mathcal{V}$ even though it is a genuine, nonempty solution stratum of the joint system.

3 The correct selection criterion

Equation (3) warns that projecting the *whole* combined variety over-counts $\mathcal{V}$. One might hope to recover $\mathcal{V}$ by keeping the cell in which the adjoined PDE-equation collapsed to $0 = 0$; Example 1 shows this is wrong, as that cell is $c = 1$ whereas $\mathcal{V} = \emptyset$. The collapse cell is the *existential* locus, not the membership locus.

The correct criterion is **dimension preservation**: $c \in \mathcal{V}$ iff the combined solution space at c still equals the *full* ansatz solution space, i.e. the PDE removed nothing. In Example 1 no cell retains the full two-dimensional space (the $c = 1$ cell keeps only a one-dimensional slice), so $\mathcal{V} = \emptyset$, consistent with the direct reduction.

4 Contrast: a one-dimensional ansatz has no partial strata

Shrink the ansatz to one dimension and test a second-order PDE:

$$A(c) : \quad \Psi' - c\Psi = 0 \quad (\Psi = Ke^{cx}), \quad P : \quad \Psi'' - \Psi = 0.$$

Now P 's leader Ψ'' dominates A 's leader Ψ' . Reducing P modulo A gives $\Psi'' \rightarrow c^2\Psi$, hence $r = (c^2 - 1)\Psi$ and

$$\mathcal{V} = V(c^2 - 1) = \{+1, -1\}.$$

The combined decomposition produces the cells

$$\{c^2 \neq 1 : \Psi \equiv 0\}, \quad \{c = 1 : \Psi = Ke^x\}, \quad \{c = -1 : \Psi = Ke^{-x}\},$$

and the nontrivial cells $c = \pm 1$ now retain the *entire* (one-dimensional) family. Here the collapse cells coincide with $\mathcal{V}$ and there are no partial strata: a one-dimensional family admits no proper nontrivial sub-family.

5 Autoreducedness does not preclude the strata: a PDE example

One might conjecture a combinatorial precondition for the partial strata: that they arise only when the reduced PDE and the ansatz still *interact* under Ritt reduction. In Example 1 the remainder $r = \Psi' - \Psi$ can reduce the ansatz equation $\Psi'' - c\Psi$ (its leader Ψ' has Ψ'' as a proper derivative), and it was exactly that rewriting which produced the split on $1 - c$. What if reduction terminates completely — r can reduce nothing in A and A nothing in r , so that $A \cup \{r\}$ is autoreduced? Does that preclude a partial-solution stratum?

In the ordinary-differential setting (one derivation, one unknown) the conjecture holds, but only vacuously. There A is a single equation with leader $\Psi^{(k)}$, and the reduced remainder involves only $\Psi, \dots, \Psi^{(k-1)}$; if r involves the unknown at all, its leader $\Psi^{(j)}$ has $\Psi^{(k)}$ as a proper derivative, so r *can* always reduce A . The hypothesis therefore forces $r = p(c)$, a pure constants condition — and then both computations return $V(p)$ carrying the full ansatz family: dimension-preserving, no strata.

With several derivations, leaders can be incomparable and the conjecture fails.

Example 2 *Two derivations ∂_x, ∂_y :*

$$A(c) : \quad \Psi_{yy} - c\Psi = 0, \quad P : \quad \Psi_x - \Psi = 0.$$

P is already reduced with respect to A (Ψ_x is not a derivative of Ψ_{yy}), so $r = P$; and r can reduce nothing in A , whose equation contains only Ψ_{yy} and Ψ , neither a derivative of Ψ_x . Thus $A \cup \{r\}$ is autoreduced — and even *coherent*: the leaders' least common derivative is Ψ_{xyy} , and the Δ -polynomial

$$\partial_y^2 r - \partial_x A = (\Psi_{xyy} - \Psi_{yy}) - (\Psi_{xyy} - c\Psi_x) = c\Psi_x - \Psi_{yy} \longrightarrow c\Psi - c\Psi = 0.$$

The combined decomposition therefore emits $A \cup \{r\}$ unchanged, as a single cell, with no splitting at all.

Membership: the projection coefficients of r are $+1$ and -1 , so $\mathcal{V} = \emptyset$ — correct, since the general ansatz solution $\Psi = F(x)e^{\omega y} + G(x)e^{-\omega y}$ ($\omega^2 = c$; F, G arbitrary functions) does not satisfy $\Psi_x = \Psi$.

Joint variety: $\Psi_x = \Psi$ forces $\Psi = e^x h(y)$, and then $h'' = c h$, so

$$\Psi = e^x (K_1 e^{\omega y} + K_2 e^{-\omega y})$$

— nontrivial at *every* c , and a proper subfamily of the ansatz family (the arbitrary functions F, G collapse to constant multiples of e^x). Every c is a partial-solution stratum; here the stratum is the whole parameter space.

The structural reading: “ $A \cup \{r\}$ is autoreduced and coherent” means the decomposition has no more algebra to do — it is precisely a finished cell — *not* that r imposes no constraint. The cell’s solution set is the joint variety, and whenever that sits strictly between 0 and $\text{Sol}(A(c))$ at some $c \notin \mathcal{V}$, a partial stratum is present. What precludes the strata is r having no non-constant part, $r = p(c)$ — which, in the single-derivation case, is the only way the autoreducedness hypothesis can be met at all. The strata are a fact of the solution geometry, not of the rewriting combinatorics.

6 Reading

The membership formulation (1) asks: *for which ODE-coefficients c is the PDE an automatic consequence of the ansatz?* — so that the ansatz family is, as a whole, a PDE-solution family. The partial strata answer a different question: *for which c does the ansatz family happen to contain a PDE solution it was not designed to?* — coincidental solutions living on a proper sub-family at special parameter values. They are real solutions, but invisible to the membership formulation by construction.

For the hydrogen computation this means: adjoining the Schrödinger equation to the ansatz and decomposing surfaces parameter values where a degenerate, lower-dimensional slice of the ansatz solves the PDE while the generic member does not. If the goal is the membership locus $\mathcal{V}$, either reduce P modulo A directly, or — having decomposed the combined system — keep only the dimension-preserving cells. Projecting the full joint variety answers the existential question and returns $\mathcal{V}$ together with the partial strata.